

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Craft and Structure	Cross-Text Connections	Medium

Question

Text 1

Microbes are tiny organisms in the soil, water, and air all around us. They thrive even in very harsh conditions. That’s why Noah Fierer and colleagues were surprised when soil samples they collected from an extremely cold, dry area in Antarctica didn’t seem to contain any life. The finding doesn’t prove that there are no microbes in that area, but the team says it does suggest that the environment severely restricts microbes’ survival.

Text 2

Microbes are found in virtually every environment on Earth. So it’s unlikely they would be completely absent from Fierer’s team’s study site, no matter how extreme the environment is. There were probably so few organisms in the samples that current technology couldn’t detect them. But since a spoonful of typical soil elsewhere might contain billions of microbes, the presence of so few in the Antarctic soil samples would show how challenging the conditions are.

Based on the texts, Fierer’s team and the author of Text 2 would most likely agree with which statement about microbes?

Answer

- A. Most microbes are better able to survive in environments with extremely dry conditions than in environments with harsh temperatures.
- B. A much higher number of microbes would probably be found if another sample of soil were taken from the Antarctic study site.
- C. Microbes are likely difficult to detect in the soil at the Antarctic study site because they tend to be smaller than microbes found in typical soil elsewhere.
- D. Most microbes are probably unable to withstand the soil conditions at the Antarctic study site.

Correct Answer: D

Rationale

Choice D is the best answer because it presents a statement about microbes with which Fierer’s team (Text 1) and the author of Text 2 would most likely agree. Text 1 states that microbes usually thrive in very harsh conditions, and so Fierer’s team was surprised when samples collected from an extremely cold and dry area of Antarctica didn’t appear to contain any life. Fierer’s team says that though this doesn’t conclusively prove there are no microbes in the area, it suggests that microbes would have a notably difficult time surviving in the environment. The author of Text 2 says it’s unlikely that there would be no microbes at all in the Antarctic study site from which Fierer’s team retrieved soil samples and that there may have been hard-to-detect microbes in the samples. However, the presence of only a few microbes in the Antarctic samples rather than the billions found in a typical soil sample (which would presumably be much easier to detect) would illustrate conditions in the Antarctic soil that make it difficult for microbes to thrive. Since Fierer’s team says that the seeming absence of microbes in the Antarctic samples suggests an unusually harsh environment and the author of Text 2 says that even if there are a few undetectable microbes in the samples, the relatively tiny number of microbes would also suggest an unusually harsh environment, then Fierer’s team and the author of Text 2 would most likely agree that most microbes are unable to withstand the soil conditions at the Antarctic study site.

Choice A is incorrect. The samples taken by Fierer’s team were from an area of Antarctica that is described in part as extremely dry, and these samples didn’t appear to have any life. Therefore, even though these samples also came from an extremely cold area, Fierer’s team wouldn’t

argue based on the evidence available that microbes were better able to survive in dry conditions than in areas with harsh temperatures. Moreover, the author of Text 2 says that microbes are found in virtually every environment on Earth but doesn't compare dry environments and harsh environments. Choice B is incorrect. Nothing in Text 1 indicates that another collection of samples from the Antarctic study site might yield different results from the samples already taken by Fierer's team. The author of Text 2 does state that microbes are found in virtually every environment on Earth and suggests that new technology may be better able to detect so few microbes in a soil sample, but the author of Text 2 concludes that the unusual absence of microbes in the Antarctic samples is evidence of the harsh Antarctic environment. Therefore, there is no reason to believe that the author of Text 2 thinks that another sample drawn from that same harsh environment would yield a much higher number of microbes. Choice C is incorrect. The author of Text 2 does speculate that there may have been so few microbes in the Antarctic samples that current technology couldn't detect them, but the author doesn't speculate that this is due to the size of the microbes. Moreover, nothing that Fierer's team says suggests that they are speculating that their samples might have microbes that are smaller than microbes in typical soil samples.